UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2007 question paper

5054 PHYSICS

5054/02 Paper 2 (Theory), maximum raw mark 75

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1 unit penalty per question, expect 2 or more sig. figs and 1 where 2nd fig is zero.
Fractions are treated as not showing final answer but can score C marks. Accept a fraction in Question 5.

Section A

1 **(a)** accelerates or speed increases **from rest/for 2-4s/for 8-20m** B1
then a constant/uniform speed **or** velocity B1

(b) between 7 and 8 m B1

(c) distance 80 (+ 2) or s= d/t in any algebraic or numerical form C1
7.3 **or** 7.27 **or** 7.273 m/s A1 [5]

2 **(a)** **(i)** **molecules** move faster or more **kinetic energy** (when hotter) B1
(more) **molecules** have (enough) energy/speed **and** escape/leave surface/ break bonds/overcome forces of attraction B1

(ii) large(r) area or wind or drier/dry atmosphere/draught or lower **atmospheric** pressure B1

(b) 40 seen or (E=) mL algebraic or numerical C1
92000 J A1 [5]

3 **(a)** mention of lower and upper fixed points **or** 0(ºC) and 100(ºC) **or** ice point/steam point B1
(marks made on) thermometer with ice/**water** mixture
and (steam above) boiling water (at atmospheric pressure) B1
divided into 100 (equal) parts (**accept** 10 parts marked 10,20 etc.) B1

(b) **(i)** 120ºC **or** –10ºC to 110ºC B1

(ii) each degree/scale marking/10ºC/division is an equal distance/0.9–1.1mm/cm/expansion
or appropriate graph a straight line B1

(c) 10ºC and 20ºC marks clearly further up thermometer **and** roughly equal spacing B1 [6]

4 **(a)** reflections correct by eye B1

(b) **all** the ray **reflects** back (into the denser medium/glass)
or reflection **and** no refraction/escape into air B1

(c) more calls **or** greater bandwidth **or** more/faster data(/sec)/information **or** better quality **or** less power loss/energy loss/attenuation **or** greater distance (between repeaters) **or** harder to tap **or** less noise/interference B1

(d) $f = v/\lambda$ in any form numerical or algebraic C1
3.3×10^{14} Hz A1 [5]

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5 **(a)** any ray from top of object correct through lens within 1mm of optical centre or F B1
other ray from same point correct through lens meeting 1st ray **and** none wrong B1

(b) **(i)** image size/object size (accept image distance/object distance **or** v/u) B1

(ii) 0.55–0.65 **ecf diagram in (a) sizes or distances** B1

(c) rays completed to retina but would meet behind retina B1 [5]

6 **(a)** at least 3 approx straight lines in middle of coil B1
one loop around top or bottom of coil B1
correct direction shown on at least one line or loop B1

(b) **(i)** 100 oscillations/cycles/waves/changes in direction (and back again) in 1 sec
or time for one oscillation etc. 0.01s B1

(ii) coil/cone moves left to right **or** back and forth **or** oscillates/vibrates **or** in and out B1
(current in) coil produces magnetic field/pole(s)
or Left Hand Rule/current in magnetic field (gives force) B1
fields/poles (of coil) oscillate/reverse B1
force (exerted by magnet) oscillates/alternates (**accept** attracts/repels) B1 ANY 2 B2 [7]

7 **(a)** **(i)** 6 Ω B1

(ii) $1/R = 1/R_1 + 1/R_2$ algebraic or numerical C1
2 Ω A1

(b) $I = V/R$ algebraic or numerical C1
6 A ecf (ii) A1

(c) (I =) 2 (A) **or** proportionality idea/potential divider idea seen C1
8 V A1 [7]

8 **(a)** different number of neutrons/ mass number B1

(b) **time** taken to halve M1
(number of) nuclei/atoms/activity/count (rate)
(**not** radioactivity/amount/mass/substance/clearly **one** nucleus/particles) A1

(c) **mark on graph** at 2000 or at two suitable points B1
12 s B1 [5]

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Section B

9 (a) (i) *A* = *B* (assume opposite direction and co-linear) B1

(ii) B > A (assume opposite direction and co-linear) B1 [2]
maximum of 1 mark if directions wrong

(b) towards **centre** of circle/corner B1 [1]

(c) (i) 0 **and** 8–9 s B1

(ii) chemical (potential energy) (**accept** electrical if electrical car clear) B1
to kinetic energy **or** K.E. increases B1
thermal energy/heat/internal energy produced B1
Max 2/3 if clear error

(iii) acceleration = (*v*–*u*)/t **or** gradient C1
(7.8 to) 8/5 (**accept** any corresponding period e.g. 8s 12.6–12.8, 6s 9.4–9.6) C1
1.6 m/s^2 (**accept** 1.56–1.60) A1

(iv) area under graph **or** average speed 4 (m/s) **or** ½ 5 speed used in **(iii)** at 5 s C1
½ x (7.8 to) 8 x 5 C1
20 m (**accept** 19.5–20; **ecf speed used in (iii)** at 5 s) A1 [10]

(d) speed of car/friction with road (accept slippery road or ice or water or oil on road)/ friction in engine/tyre condition or area or pressure/air resistance/wind speed or direction/mass or inertia **of car or passengers**/slope of road B2 [2]

10 (a) resistance of cables B1
power/energy/heat loss **or** voltage drop **or** current low in cables/wires clear B1 [2]

(b) low(er) current in line **or** less voltage drop/power/heat/energy loss B1
A steps voltage up or increases voltage or reduces current B1
B steps voltage down or decreases voltage or increases current B1 [3]

(c) (i) two coils (no label needed) M1
coils labelled/described primary/input **and** secondary/output
or insulated **or** copper A1
coils on complete (soft) iron (core) B1
(accept from labelled diagram or description)

(ii) alternating/changing **current** input B1
(alternating) magnetic field (produced in core **or** coil) B1
induced e.m.f./voltage/current (in secondary coil) B1 [6]

(d) (i) I = P/V algebraic or numerical C1
3 A A1

(ii) E = VIt **or** Pt algebraic or numerical **or** 600 (s) used C1
414 000 (J) **or** 414kJ **or** 410 000 (J) A1 [4]

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11 (a) –ve charge/electrons moves or rod gains electrons C1
–ve charge/electrons move from cloth to rod A1
positive electrons scores 0/2 in **(a)** and **(b)(i)** +ve moves max 1 mark

(b) (i) (electrons) move to right/to X/to opposite side (to rod) / B1
(electrons or –ve) repelled (by rod) or like charges repel B1

(ii) +ve on left and –ve on right, inside or outside sphere B1

(iii) +ve attracted to rod or unlike charges or +ve and –ve attract B1
repulsion of –ve on sphere (by rod) weaker (than attraction) B1 [7]

(c) (i) connection of sphere to earth/ground/0 V B1

(ii) move down to the ground/earth **or** electrons on right/at X removed B1
repelled (by –ve on rod) **or** move from –ve to 0 potential B1

(iii) only +ve on sphere at left or clearly more positive on left than on right B1 [4]

(d) Sensible example of a **use** of charging,
e.g. precipitator, photocopier, spray painting, gold leaf electroscope, plates in CRO M1
ink jet printer, Van de Graff generator, piezoelectric devices, capacitor, lightning conductor
simple diagram showing effect A1
a correctly charged object clear A1
description of the function that the charge performs A1 [4]

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